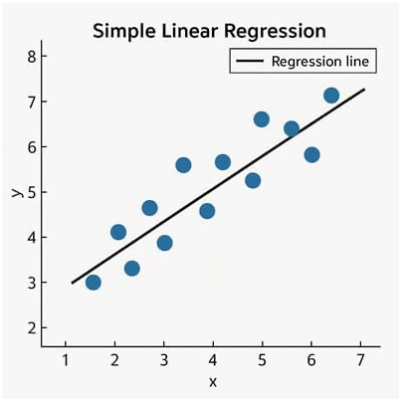
ISY503 - Module 3 – Machine Learning Models

Learning Activity 1

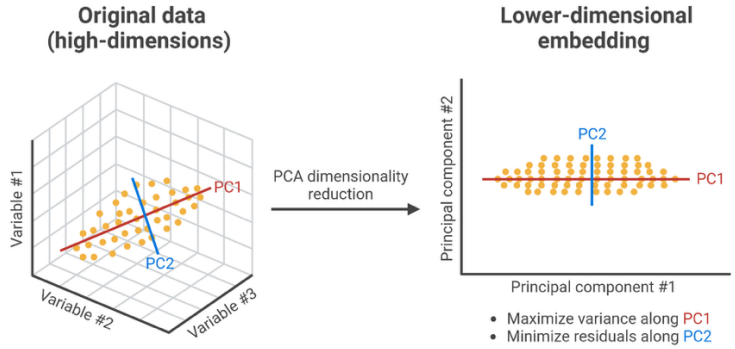
My research on 4 ML models:

Linear Regression:



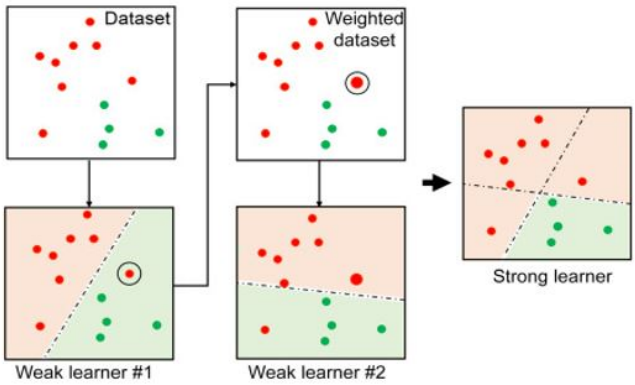
predicts a continuous numeric output as a weighted sum of input features; fits a line/plane by minimizing mean squared error. Supervised regression algorithm, with interpretable coefficients, fast to train that assumes linear relationship.

Principal Component Analysis (PCA):



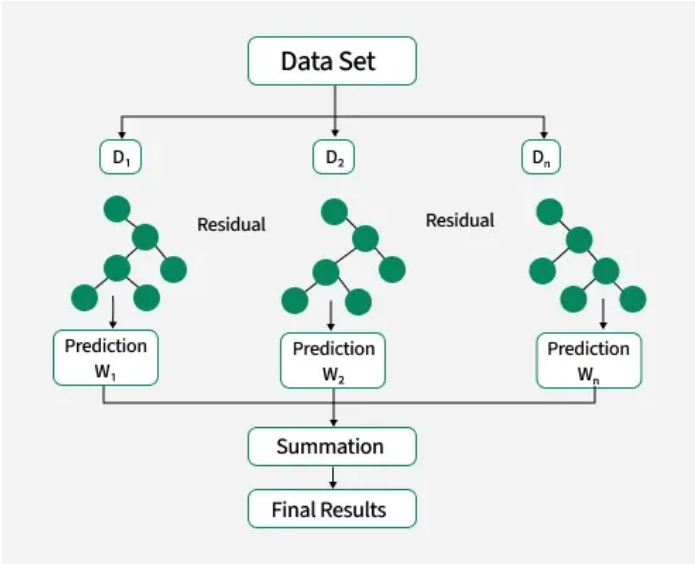
Unsupervised dimensionality reduction technique that projects data onto orthogonal axes (principal components) ordered by explained variance. Reduces noise, removes correlated features, and compresses data – not a classifier but a preprocessing step.

AdaBoost (Adaptive Boosting):



Sequential ensemble that trains weak learners (typically shallow trees) where each iteration reweights misclassified samples so the next learner focuses on harder cases. Final prediction is a weighted vote of all learnes; sensitive to outliers and noise.

XGBoost (Extreme Gradient Boosting):



Highly optimized gradient boosting framework that builds trees sequentially to minimize a regularized objective (L1/L2). Handles missing values natively, supports parallelization, and consistently dominates tabular ML benchmarks.